

MUONIO ALAMUONIO, Finland

WMO: 028230

Lat: 67.9583N

Lon: 23.6847E

Elev: 253

StdP: 98.33

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.4	-36.4	0.1	-33.5	-32.9	0.2	-30.4	8.3	-4.6	7.4	-4.7	0.7	330	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	24.1	16.1	21.9	15.1	19.8	14.2	17.3	21.8	16.1	20.4	14.9	18.6	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.4	18.7	14.3	10.5	17.2	13.2	9.8	16.4	49.5	21.8	45.8	20.5	42.5	18.7	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.2	5.4	-36.1	26.2	4.1	1.8	-39.0	27.5	-41.4	28.5	-43.7	29.5	-46.7	30.8		
(5)			-36.2	18.6	4.1	1.4	-39.1	19.6	-41.5	20.4	-43.8	21.2	-46.7	22.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-13.9	-13.2	-8.0	-1.5	4.9	10.8	14.3	11.7	6.5	-0.6	-7.7	-11.6	(6)	
	DBStd	11.31	8.09	8.02	6.15	4.50	4.27	3.83	3.28	3.43	3.69	4.79	6.78	8.08	(7)	
	HDD10.0	4168	740	651	558	345	169	35	3	21	115	329	532	669	(8)	
	HDD18.3	6928	998	884	817	595	417	228	132	206	354	588	782	927	(9)	
	CDD10.0	291	0	0	0	0	10	59	137	74	11	0	0	0	(10)	
	CDD18.3	9	0	0	0	0	0	2	6	1	0	0	0	0	(11)	
	CDH23.3	87	0	0	0	0	4	16	57	11	0	0	0	0	(12)	
	CDH26.7	7	0	0	0	0	0	1	6	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.2	1.7	2.1	2.5	2.3	2.7	2.7	2.4	2.2	2.3	2.1	1.8	1.8	(14)	
(15) Precipitation	PrecAvg	563	36	31	28	30	41	63	77	73	51	48	45	39	(15)	
	PrecMax	694	77	60	66	73	87	116	128	127	103	105	83	76	(16)	
	PrecMin	412	6	6	8	7	8	9	25	32	11	11	14	9	(17)	
	PrecStd	81	15	14	15	14	21	27	28	27	26	22	19	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.3	5.8	12.5	22.9	25.5	27.2	24.8	17.4	10.1	5.0	3.2	(19)	
		MCWB	0.7	1.5	2.5	7.0	14.0	16.5	17.4	17.1	12.2	8.5	4.5	1.5	(20)	
	2%	DB	-0.2	0.5	3.5	9.0	18.7	22.3	24.9	22.1	15.2	8.7	2.4	0.9	(21)	
		MCWB	-0.6	-0.5	0.8	4.8	11.5	14.4	16.7	15.9	11.6	7.5	1.9	0.4	(22)	
	5%	DB	-1.8	-0.9	1.6	6.9	14.6	19.9	22.7	19.8	13.7	7.2	1.1	-0.1	(23)	
		MCWB	-2.2	-1.4	-0.3	3.3	9.0	13.4	15.8	14.7	11.0	6.2	0.8	-0.4	(24)	
	10%	DB	-3.3	-2.4	0.1	5.0	11.9	17.6	20.4	17.6	12.3	5.5	0.2	-1.1	(25)	
		MCWB	-3.7	-2.9	-1.3	2.1	7.3	12.0	14.9	13.7	10.4	4.6	0.0	-1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.7	2.6	7.3	14.3	17.6	19.1	18.5	14.0	9.4	4.8	1.7	(27)	
		MCDB	1.4	3.0	5.4	12.1	21.6	22.9	23.8	22.6	16.0	9.7	4.9	2.7	(28)	
	2%	WB	-0.6	-0.2	1.1	4.9	11.7	15.4	17.7	16.6	12.6	7.7	2.0	0.4	(29)	
		MCDB	-0.3	0.4	3.0	8.4	17.6	20.5	22.5	20.5	14.1	8.2	2.4	0.8	(30)	
	5%	WB	-2.1	-1.4	0.0	3.6	9.6	13.9	16.6	15.3	11.6	6.2	0.9	-0.3	(31)	
		MCDB	-1.8	-0.8	1.3	6.4	13.8	18.5	21.0	18.5	13.1	7.1	1.0	0.0	(32)	
	10%	WB	-3.7	-2.9	-1.2	2.4	7.9	12.6	15.5	14.2	10.6	4.7	0.1	-1.5	(33)	
		MCDB	-3.4	-2.4	0.0	4.6	11.3	16.9	19.7	17.0	12.1	5.5	0.2	-1.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.2	10.3	9.3	8.3	7.9	8.4	8.6	7.3	5.5	6.3	7.6	(35)	
		MCDBR	7.1	6.6	9.0	10.7	13.2	11.6	12.4	11.9	9.2	5.9	4.5	5.4	(36)	
		MCWBR	7.1	5.9	7.0	7.0	7.3	5.8	5.7	6.1	5.9	4.8	4.3	5.2	(37)	
	5% WB	MCDBR	7.0	6.6	7.6	9.6	11.7	10.0	10.2	9.8	7.4	5.2	4.2	5.4	(38)	
		MCWBR	6.9	5.9	6.2	6.3	6.9	5.4	5.4	5.7	5.1	4.6	4.1	5.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.188	0.248	0.284	0.314	0.333	0.337	0.346	0.331	0.300	0.267	0.351	N/A	(40)	
	taud		1.919	2.089	2.181	2.210	2.413	2.483	2.472	2.497	2.557	2.324	2.760	N/A	(41)	
	Ebn at Noon		111	550	751	831	861	867	845	820	763	546	159	0	(42)	
	Edh at Noon		9	48	79	103	95	91	90	78	57	38	12	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.04	0.51	2.02	3.97	4.90	5.00	4.68	3.38	1.81	0.60	0.10	0.00	(44)	
	RadStd		0.00	0.06	0.12	0.18	0.48	0.35	0.46	0.25	0.21	0.10	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page